Unconfounded but inflated causal estimates

Mathematical description

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I Setting

I. 1 Objectives

This document aims to provide a mathematical derivation to illustrate the trade-off underlined in this paper. Such a derivation may provide some intuition regarding the various mechanisms at play. It does not aim in any way to provide a general proof but merely a specific one, tied to the setting considered. While we keep this setting as general as possible, as in most cases, we have to make certain assumptions, in particular regarding the data generating process considered.

The objective here is to compare the exaggeration ratio for a naive regression with omitted variable bias (OVB) to that of an Instrumental Variable (IV) model in which we instrument the variable affected by the omitted variable. We expect the naive regression to be biased but more precise than the IV and therefore less subject to exaggeration problems. Roughly, we aim to derive the conditions under which the exaggeration due to the loss in precision is greater than the OVB. Thus, the objective is to compare the following quantities:

$$\mathbb{E}\left[\hat{\beta}_{\text{OVB}} |\ \beta_1, b_{\text{OVB}}, \sigma_{\text{OVB}}, \hat{\beta}_{\text{OVB}} > z_\alpha \sigma_{\text{OVB}}\right]$$
$$\mathbb{E}\left[\hat{\beta}_{\text{IV}} |\ \beta_1, \sigma_{\text{IV}}, \hat{\beta}_{\text{IV}} > z_\alpha \sigma_{\text{IV}}\right]$$

where OVB and IV denote the naive regression and the IV respectively. β_1 represents the true effect of interest, $\hat{\beta}_{..}$ its estimate, $\sigma_{..}$ the standard error of the estimate and z_α the threshold value of a two-sided test of size α.

To compute these values, we need to know the distribution of the estimates. After describing the data generating process, we thus derive the asymptotic distribution of these estimates. We then compute and compare these quantities as a function of the parameters of interests. The main parameter of interest is linked the proportion of the variance in the explanatory variable of interest retained for identification.

I. 2 Data generating process

Consider the usual linear regression model to which we add omitted variables. We limit the number of covariates considered to a constant, an observed variable and an omitted variable. For all individual $i \in 1, ..., n$, we make the following assumptions:

$$y_i = \mathrm{x}_i' \boldsymbol{\beta} + \delta w_i + u_i \tag{1}$$

where $\mathrm{x}_i \in \mathbb{R}^2$ is a vector including a constant and the observed explanatory variable such that $\mathrm{x}_i = (1, x_i)'$, $y_i \in \mathbb{R}$ the observed outcome of interest, $w_i \in \mathbb{R}$ an unobserved omitted variable, $u_i \in \mathbb{R}$

an unobserved error term. $\boldsymbol{\beta} \in \mathbb{R}^2$ and $\delta \in \mathbb{R}$ are unknown parameters. $\boldsymbol{\beta} = (\beta_0, \beta_1)'$ and β_1 is the parameter of interest. For readability, we introduce the usual vector notation such that for instance $y = (y_1, ..., y_n)'$.

We further assume that the usual OLS assumptions hold: $(y_i, \mathrm{x}_i, w_i, u_i)$ are iid, finite second moments, *ie* $\mathbb{E}[y_i^2] < \infty$, $\mathbb{E}[w_i^2] < \infty$, $\mathbb{E}[x_i^2] < \infty$, positive-definiteness of $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']$, u_i is conditional mean-zero, *ie* $\mathbb{E}[u_i|x_i, w_i] = 0$, and is uncorrelated with x_i and w_i, *ie* $\mathbb{E}[x_i u_i] = 0$ and $\mathbb{E}[w_i u_i] = 0$. Note that conditional mean-zero implies that $\mathbb{E}[u_i|x_i] = 0$ (since $\mathbb{E}[u_i|x_i] = \mathbb{E}[\mathbb{E}[u_i|x_i, w_i|x_i]] = \mathbb{E}[0|x_i] = 0$). We further assume homoskedasticity such that $\mathbb{E}[u_i^2|x_i, w_i] = \sigma_u^2$ is constant. It implies that $\mathbb{E}[u_i^2|x_i] = \sigma_u^2$ (since $\mathbb{E}[u_i^2|x_i] = \mathbb{E}[\mathbb{E}[u_i^2|x_i, w_i]|x_i] = \mathbb{E}[\sigma_u^2|x_i] = \sigma_u^2$).

We however assume that w_i is unobserved, correlated with x_i and that $\delta \neq 0$. To simplify the derivations, we further assume that the unobserved variable is centred, *ie* $\mathbb{E}[w_i] = 0$. We also assume that the variance of the component of w_i that is orthogonal to x_i (denoted $w_i^\perp$) does not vary with x_i. Thus, $\mathrm{Var}(w_i^\perp|x_i) = \mathrm{Var}(w_i^\perp)$.

We further assume that there exists a valid instrumental variable z_i for x_i. Let $\mathrm{z} = (1, z_i)' \in \mathbb{R}^2$. Note that the model is just-identified. Since the instrument is valid, it satisfies exogeneity, *ie* $\mathbb{E}[\mathrm{z}_i u_i] = 0$ and $\mathbb{E}[z_i w_i] = 0$, relevance, *ie* $\mathrm{rank}(\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']) = 2$, and positive-definiteness of $\mathbb{E}[\mathrm{z}_i\mathrm{z}_i']$. To simplify interpretation, we further assume a data generating process for x_i:

$$x_i = \eta_0 + \eta_1 z_i + \gamma w_i + \epsilon_i \tag{2}$$

where $(\eta_0, \eta_1, \gamma) \in \mathbb{R}^3$ are unknown parameters. We assume that ϵ_i is uncorrelated with z_i and w_i, *ie* $\mathbb{E}[z_i\epsilon_i] = 0$ and $\mathbb{E}[w_i\epsilon_i] = 0$. Note that, since x_i and w_i are correlated, $\gamma \neq 0$.

The following Directed Acyclic Graph (DAG) sums up the relationships between the variables. It also helps underline the independence, or not, of some variables.

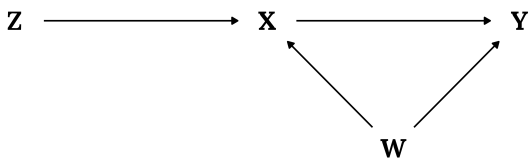

Figure 1 – DAG of the data generating process

I. 3 Summary of the key assumptions

To ease the use of assumptions in the following sections, we summarize some that we will explicitly mobilize:

- $y_i = \mathrm{x}_i'\boldsymbol{\beta} + \delta w_i + u_i$
- $x_i = \eta_0 + \eta_1 z_i + \gamma w_i + \epsilon_i$
- $\mathbb{E}[u_i|x_i, w_i] = 0$ and $\mathbb{E}[u_i|x_i] = 0$
- $\mathbb{E}[u_i^2|x_i, w_i] = \sigma_u^2$ and $\mathbb{E}[u_i^2|x_i] = \sigma_u^2$
- $\mathbb{E}[w_i] = 0$
- $\mathbb{E}[z_i u_i] = 0$ and $\mathbb{E}[z_i w_i] = 0$
- $\mathbb{E}[z_i\epsilon_i] = 0$ and $\mathbb{E}[w_i\epsilon_i] = 0$
- $\mathrm{Var}(w_i^\perp|x_i) = \mathrm{Var}(w_i^\perp)$

II Asymptotic distribution of the estimates

As discussed above, the goal is to compare the exaggeration ration in the case of the naive regression to that of the corresponding reduced form and IV. In this section, we **derive the asymptotic distributions of these estimates**.

II. 1 Naive regression (OVB)

The true DGP is given by equation 1. Since, we do not observe W, we consider the projection of y on X only:

$$y = X\boldsymbol{\beta}_{\text{OVB}} + v \tag{3}$$

where $v = (v_1, ..., v_n)' \in \mathbb{R}^n$ and by definition of the projection, $\mathbb{E}[X'v] = 0$.

II. 1.1 Bias of the estimator

From equation 3 we get:

$$\begin{aligned}
& X'y = X'X\boldsymbol{\beta}_{\text{OVB}} + X'v \\
\Rightarrow\quad & \mathbb{E}[X'y] = \underbrace{\mathbb{E}[X'X]}_{\text{pos. def.}}\boldsymbol{\beta}_{\text{OVB}} + \underbrace{\mathbb{E}[X'v]}_{0} \\
\Leftrightarrow\quad & \boldsymbol{\beta}_{\text{OVB}} = \mathbb{E}[X'X]^{-1}\mathbb{E}[X'y] \\
\Leftrightarrow\quad & \boldsymbol{\beta}_{\text{OVB}} = \mathbb{E}[X'X]^{-1}\mathbb{E}[X'(X\boldsymbol{\beta} + \delta w + u)] && \text{cf eq. 1} \\
\Leftrightarrow\quad & \boldsymbol{\beta}_{\text{OVB}} = \boldsymbol{\beta} + \mathbb{E}[X'X]^{-1}\mathbb{E}[X'w]\delta + \underbrace{\mathbb{E}[X'u]}_{0} \\
\Leftrightarrow\quad & \boldsymbol{\beta}_{\text{OVB}} = \boldsymbol{\beta} + \mathbb{E}[X'X]^{-1}\mathbb{E}[X'w]\delta
\end{aligned}$$

Since X and w are correlated and $\delta \neq 0$, $\boxed{\boldsymbol{\beta}_{\text{OVB}} \neq \boldsymbol{\beta}}$. $\mathbb{E}[X'X]^{-1}\mathbb{E}[X'w]$ is the coefficient matrix of the projection of w on X. A similar derivation and discussion is carried out in Hansen (2022) for instance.

Let $\hat{\boldsymbol{\beta}}_{\text{OVB}}$ denote the least square estimator of $\boldsymbol{\beta}_{\text{OVB}}$. We thus have:

$$\hat{\boldsymbol{\beta}}_{\text{OVB}} = (X'X)^{-1}X'y \tag{4}$$

$$\begin{aligned}
\Rightarrow\quad & \hat{\boldsymbol{\beta}}_{\text{OVB}} = (X'X)^{-1}X'(X\boldsymbol{\beta}_{\text{OVB}} + v) && \text{cf eq. 3} \\
\Rightarrow\quad & \mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{OVB}} \mid X] = \boldsymbol{\beta}_{\text{OVB}} + (X'X)^{-1}\underbrace{\mathbb{E}[X'v]}_{0} \\
\Rightarrow\quad & \mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{OVB}}] = \mathbb{E}[\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{OVB}} \mid X]] = \boldsymbol{\beta}_{\text{OVB}} && \text{by the Law of Iterated Expect. (LIE)}
\end{aligned}$$

Thus, <u>$\hat{\boldsymbol{\beta}}_{\text{OVB}}$ is an unbiased estimator of $\boldsymbol{\beta}_{\text{OVB}}$ but a biased estimator of $\boldsymbol{\beta}$</u>. We then compute its asymptotic distribution.

II. 1.2 Asymptotic distribution

We can write:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{OVB}} - \boldsymbol{\beta}_{\text{OVB}}) = \left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i\mathrm{x}_i'\right)^{-1} \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i v_i\right)$$

By the Weak Law of Large Numbers (WLLN), $\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i\mathrm{x}_i' \xrightarrow{p} \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']$.
Since $\mathbb{E}[\mathrm{x}_i v_i] = 0$, by the Central Limit Theorem (CLT), $\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i v_i\right) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2])$.
Applying Slutsky's theorem then yields:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{OVB}} - \boldsymbol{\beta}_{\text{OVB}}) \xrightarrow{d} \mathcal{N}\left(0, \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2]\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\right)$$

And thus,

$$\boxed{\hat{\boldsymbol{\beta}}_{\text{OVB}} \xrightarrow{d} \mathcal{N}\left(\boldsymbol{\beta} + \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i w_i]\delta, \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2]\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}/n\right)} \tag{5}$$

We are interested in the second component of $\hat{\boldsymbol{\beta}}_{\text{OVB}}$. To retrieve it we need to compute $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}$, $\mathbb{E}[\mathrm{x}_i w_i]$ and $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2]$.

$$\mathbb{E}[\mathrm{x}_i\mathrm{x}_i'] = \mathbb{E}\begin{bmatrix} 1 & x \\ x & x^2 \end{bmatrix} = \begin{bmatrix} 1 & \mu_x \\ \mu_x & \sigma_x^2 + \mu_x^2 \end{bmatrix} \quad \Rightarrow \quad \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1} = \frac{1}{\sigma_x^2}\begin{bmatrix} \sigma_x^2 + \mu_x^2 & -\mu_x \\ -\mu_x & 1 \end{bmatrix}$$

$$\mathbb{E}[\mathrm{x}_i w_i] = \mathbb{E}\begin{bmatrix} w_i \\ x_i w_i \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbb{E}[x_i]\underbrace{\mathbb{E}[w_i]}_{0} + \mathrm{cov}(x_i, w_i) \end{bmatrix} = \begin{bmatrix} 0 \\ \eta_1 \underbrace{\mathrm{cov}(z_i, w_i)}_{0} + \gamma \underbrace{\mathrm{var}(w_i)}_{\sigma_w^2} + \underbrace{\mathrm{cov}(\epsilon_i, w_i)}_{0} \end{bmatrix} = \begin{bmatrix} 0 \\ \gamma\sigma_w^2 \end{bmatrix}$$

$$\Rightarrow \quad \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i w_i] = \frac{\gamma\sigma_w^2}{\sigma_x^2}\begin{bmatrix} -\mu_x \\ 1 \end{bmatrix} \tag{6}$$

Note that $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2] \overset{\text{LIE}}{=} \mathbb{E}[\mathrm{x}_i\mathrm{x}_i'\mathbb{E}[v_i^2|\mathrm{x}_i]]$. We thus first compute $\mathbb{E}[v_i^2|\mathrm{x}_i]$, noting that:

$$\begin{aligned}
v_i &= y_i - \mathrm{x}_i'\boldsymbol{\beta}_{\text{OVB}} \\
&= \delta w_i + u_i + \mathrm{x}_i'(\boldsymbol{\beta} - \boldsymbol{\beta}_{\text{OVB}}) \\
&= \delta w_i + u_i - \underbrace{\mathrm{x}_i'\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i w_i]}_{\text{projection of } w_i \text{ on } x_i}\delta \\
&= u_i + \delta\underbrace{\left(w_i - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)\right)}_{\text{part of } w_i \text{ orthogonal to } x_i} \\
&= u_i + \delta w_i^{\perp} \qquad\qquad \text{where } w_i^{\perp} = w_i - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)
\end{aligned}$$

And thus,

$$\begin{aligned}
\mathbb{E}[v_i^2|\mathrm{x}_i] &= \mathbb{E}[(u_i + \delta w_i^{\perp})^2|x_i] \\
&= \mathbb{E}[u_i^2|x_i] + 2\delta\mathbb{E}[u_i w_i^{\perp}|x_i] + \delta^2\mathbb{E}[(w_i^{\perp})^2|x_i] \\
&= \sigma_u^2 + 2\delta\left(\mathbb{E}[u_i w_i|x_i] - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)\underbrace{\mathbb{E}[u_i|x_i]}_{0}\right) + \delta^2\mathbb{E}[(w_i^{\perp})^2|x_i] \\
&\overset{\text{LIE}}{=} \sigma_u^2 + 2\delta\mathbb{E}[w_i\underbrace{\mathbb{E}[u_i|x_i, w_i]}_{0}|x_i] + \delta^2\mathbb{E}[(w_i^{\perp})^2|x_i] \\
&= \sigma_u^2 + \delta^2\mathbb{E}[(w_i^{\perp})^2|x_i]
\end{aligned}$$

Notice that, by the law of total variance, $\mathbb{E}[(w_i^{\perp})^2|x_i] = \mathrm{Var}(w_i^{\perp}|x_i) + \mathbb{E}[w_i^{\perp}|x_i]^2$. Now, since $w_i^{\perp}$ is the component of w_i that is orthogonal to x_i and by the projection interpretation of the conditional variance, $\mathbb{E}[w_i^{\perp}|x_i] = 0$. And thus, since by assumption $\mathrm{Var}(w_i^{\perp}|x_i) = \mathrm{Var}(w_i^{\perp})$,

$$\begin{aligned}
\mathbb{E}[(w_i^{\perp})^2|x_i] &= \mathrm{Var}(w_i^{\perp}|x_i) \\
&= \mathrm{Var}(w_i^{\perp}) \\
&= \mathbb{E}[(w_i^{\perp})^2] - \mathbb{E}[w_i^{\perp}]^2 \\
&= \mathbb{E}\left[\left(w_i - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)\right)^2\right] - \left(\underbrace{\mathbb{E}[w_i]}_{0} + \frac{\gamma\sigma_w^2}{\sigma_x^2}\underbrace{\mathbb{E}[x_i - \mu_x]}_{0}\right)^2 \\
&= \underbrace{\mathbb{E}[w_i^2]}_{\sigma_w^2} - 2\frac{\gamma\sigma_w^2}{\sigma_x^2}\left(\underbrace{\mathbb{E}[x_i w_i]}_{\gamma\sigma_w^2} - \mu_x\underbrace{\mathbb{E}[w_i]}_{0}\right) + \frac{\gamma^2\sigma_w^4}{\sigma_x^4}\underbrace{\mathbb{E}[(x_i - \mu_x)^2]}_{\sigma_x^2} \\
&= \sigma_w^2 - 2\frac{\gamma^2\sigma_w^4}{\sigma_x^2} + \frac{\gamma^2\sigma_w^4}{\sigma_x^2} \\
&= \sigma_w^2 - \frac{\gamma^2\sigma_w^4}{\sigma_x^2} \\
&= \sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)
\end{aligned}$$

And therefore,

$$\mathbb{E}[v_i^2|\mathrm{x}_i] = \sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right) \tag{7}$$

Thus, under our set of assumptions, $\mathbb{E}[v_i^2|\mathrm{x}_i]$ does not depend on x_i and $\mathbb{E}[v_i^2|\mathrm{x}_i] = \mathbb{E}[v_i^2]$. We denote this quantity σ_v^2.

We can now compute the variance of the estimator $\hat{\boldsymbol{\beta}}_{\text{OVB}}$, noting that $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' v_i^2] = \mathbb{E}[\mathrm{x}_i\mathrm{x}_i'\mathbb{E}[v_i^2|\mathrm{x}_i]] = \mathbb{E}[\mathrm{x}_i\mathrm{x}_i'\sigma_v^2] = \sigma_v^2\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']$.

Thus, plugin this and equation 6 into equation 5, for $\hat{\beta}_{\text{OVB}}$, the second component of $\hat{\boldsymbol{\beta}}_{\text{OVB}}$, we have:

$$\hat{\beta}_{\text{OVB}} \xrightarrow{d} \mathcal{N}\left(\beta_1 + \frac{\delta\gamma\sigma_w^2}{\sigma_x^2},\ \frac{\sigma_v^2}{n\ \sigma_x^2}\right) \tag{8}$$

And substituting for σ_v^2,

$$\boxed{\hat{\beta}_{\text{OVB}} \xrightarrow{d} \mathcal{N}\left(\beta_1 + \frac{\delta\gamma\sigma_w^2}{\sigma_x^2},\ \frac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{n\ \sigma_x^2}\right)} \tag{9}$$

Using the notations introduced in section I. 1, we have:

$$\begin{aligned} b_{\text{OVB}} &= \frac{\delta\gamma\sigma_w^2}{\sigma_x^2} \\ \sigma_{\text{OVB}}^2 &= \frac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{n\ \sigma_x^2} \end{aligned}$$

II. 2 Instrumental Variable

The true DGP for y is given by equation 1. Here, we instrument x_i with z_i, or more precisely $\mathrm{x}_i = (1, x_i)'$ with $\mathrm{z}_i = (1, z_i)'$. We are therefore in a just-identified case and $\hat{\boldsymbol{\beta}}_{\text{2SLS}} = \hat{\boldsymbol{\beta}}_{\text{IV}}$. And:

$$\begin{aligned} \hat{\boldsymbol{\beta}}_{\text{IV}} &= (Z'X)^{-1}Z'y \\ &= (Z'X)^{-1}Z'(X\boldsymbol{\beta} + \delta w + u) \\ &= \boldsymbol{\beta} + (Z'X)^{-1}Z'w\delta + (Z'X)^{-1}Z'u \end{aligned}$$

Thus,

$$\mathbb{E}[\hat{\boldsymbol{\beta}}_{\text{IV}}|X, Z] = \boldsymbol{\beta} + (Z'X)^{-1}\underbrace{\mathbb{E}[Z'w]}_{0}\delta + (Z'X)^{-1}\underbrace{\mathbb{E}[Z'u]}_{0} = \boldsymbol{\beta}$$

$\hat{\boldsymbol{\beta}}_{\text{IV}}$ is therefore an unbiased estimator of $\boldsymbol{\beta}$. We then derive its asymptotic distribution, noting that:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{IV}} - \boldsymbol{\beta}) = \left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{z}_i\mathrm{x}_i'\right)^{-1}\delta\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{z}_i w_i\right) + \left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{z}_i\mathrm{x}_i'\right)^{-1}\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{z}_i u_i\right)$$

The WLLN gives $\frac{1}{n}\sum_{i=1}^{n}\mathrm{z}_i\mathrm{x}_i' \xrightarrow{p} \mathbb{E}[\mathrm{z}_i\mathrm{x}_i']$ and since $\mathbb{E}[\mathrm{z}_i w_i] = 0$ and $\mathbb{E}[\mathrm{z}_i u_i] = 0$, the Central Limit Theorem (CLT) gives $\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{z}_i w_i\right) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{z}_i\mathrm{z}_i' w_i^2])$ and $\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\mathrm{z}_i u_i\right) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathrm{z}_i\mathrm{z}_i' u_i^2])$.

Now, $\mathbb{E}[\mathrm{z}_i\mathrm{z}_i' u_i^2] = \mathbb{E}[\mathrm{z}_i\mathrm{z}_i'\mathbb{E}[u_i^2|\mathrm{z}_i]] = \mathbb{E}[\mathrm{z}_i\mathrm{z}_i'\mathbb{E}(\mathbb{E}[u_i^2|\mathrm{x}_i]|\mathrm{z}_i)] = \mathbb{E}[\mathrm{z}_i\mathrm{z}_i'\mathbb{E}[\sigma_u^2|\mathrm{z}_i]] = \mathbb{E}[\mathrm{z}_i\mathrm{z}_i'[\sigma_u^2] = \sigma_u^2\mathbb{E}[\mathrm{z}_i\mathrm{z}_i']$. Now, since z_i is a valid instrument, it makes sense to have $\mathbb{E}[w_i^2|\mathrm{z}_i] = \sigma_w^2$. Thus, $\mathbb{E}[\mathrm{z}_i\mathrm{z}_i' w_i^2] = \mathbb{E}[\mathrm{z}_i\mathrm{z}_i'\mathbb{E}[w_i^2|\mathrm{z}_i]] = \sigma_w^2\mathbb{E}[\mathrm{z}_i\mathrm{z}_i']$.

Applying Slutsky's theorem then yields:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{IV}} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}\left(0, (\delta^2\sigma_w^2 + \sigma_u^2)\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{z}_i\mathrm{z}_i']\left(\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\right)'\right)$$

And thus,

$$\boxed{\hat{\boldsymbol{\beta}}_{\text{IV}} \xrightarrow{d} \mathcal{N}\left(\boldsymbol{\beta}, \frac{\delta^2\sigma_w^2 + \sigma_u^2}{n}\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{z}_i\mathrm{z}_i']\left(\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\right)'\right)}$$

We are interested in the second component of $\hat{\boldsymbol{\beta}}_{\text{IV}}$. To retrieve it we need to compute $\mathbb{E}[\mathrm{z}_i\mathrm{z}_i]$, $\mathbb{E}[\mathrm{x}_i\mathrm{z}_i']^{-1}$ and its transpose.

$$\mathbb{E}[\mathrm{z}_i\mathrm{z}_i'] = \mathbb{E}\begin{bmatrix} 1 & z \\ z & z^2 \end{bmatrix} = \begin{bmatrix} 1 & \mu_z \\ \mu_z & \sigma_z^2 + \mu_z^2 \end{bmatrix}$$

$$\mathbb{E}[\mathrm{z}_i\mathrm{x}_i'] = \begin{bmatrix} 1 & \mathbb{E}[x_i] \\ \mathbb{E}[z_i] & \mathbb{E}[z_i x_i] \end{bmatrix} = \begin{bmatrix} 1 & \eta_0 + \eta_1\mathbb{E}[z_i] + \gamma\underbrace{\mathbb{E}[w_i]}_{0} + \underbrace{\mathbb{E}[\epsilon_i]}_{0} \\ \mu_z & \eta_0\mathbb{E}[z_i] + \eta_1\mathbb{E}[z_i^2] + \gamma\underbrace{\mathbb{E}[z_i w_i]}_{0} + \underbrace{\mathbb{E}[z_i\epsilon_i]}_{0} \end{bmatrix} = \begin{bmatrix} 1 & \eta_0 + \eta_1\mu_z \\ \mu_z & \eta_0\mu_z + \eta_1(\sigma_z^2 + \mu_z^2) \end{bmatrix}$$

$$\Rightarrow \quad \mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1} = \frac{1}{\eta_1\sigma_z^2}\begin{bmatrix} \eta_0\mu_z + \eta_1(\sigma_z^2 + \mu_z^2) & -\eta_0 - \eta_1\mu_z \\ -\mu_z & 1 \end{bmatrix}$$

$$\text{and} \quad \left(\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\right)' = \frac{1}{\eta_1\sigma_z^2}\begin{bmatrix} \eta_0\mu_z + \eta_1(\sigma_z^2 + \mu_z^2) & -\mu_z \\ -\eta_0 - \eta_1\mu_z & 1 \end{bmatrix}$$

Thus,

$$\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{z}_i\mathrm{z}_i'] = \frac{1}{\eta_1\sigma_z^2}\begin{bmatrix} \eta_1\sigma_z^2 & -\eta_0\sigma_z^2 \\ 0 & \sigma_z^2 \end{bmatrix} = \begin{bmatrix} 1 & -\frac{\eta_0}{\eta_1} \\ 0 & \frac{1}{\eta} \end{bmatrix}$$

And finally,

$$\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{z}_i\mathrm{z}_i']\left(\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\right)' = \frac{1}{\eta_1\sigma_z^2}\begin{bmatrix} 2\eta_0\mu_z + \eta_1(\sigma_z^2 + \mu_z^2) + \frac{\eta_0^2}{\eta_1} & -\mu_z - \frac{\eta_0}{\eta_1} \\ -\mu_z - \frac{\eta_0}{\eta_1} & \frac{1}{\eta_1} \end{bmatrix}$$

Thus, for $\hat{\beta}_{\text{IV}}$, the second component of $\hat{\boldsymbol{\beta}}_{\text{IV}}$, we have:

$$\boxed{\hat{\beta}_{\text{IV}} \xrightarrow{d} \mathcal{N}\left(\beta_1,\ \frac{\delta^2\sigma_w^2 + \sigma_u^2}{n\ \eta_1^2\sigma_z^2}\right)} \tag{10}$$

Using the notations introduced in section I. 1, we have:

$$b_{\text{IV}} = 0 \tag{11}$$

$$\sigma_{\text{IV}}^2 = \frac{\delta^2\sigma_w^2 + \sigma_u^2}{n\ \eta_1^2\sigma_z^2} \tag{12}$$

Note that, since z_i is uncorrelated with w_i and ϵ_i,

$$\begin{aligned} & \operatorname{cov}(x_i, z_i) = \operatorname{cov}(\eta_0 + \eta_1 z_i + \gamma w_i + \epsilon_i, z_i) = \eta_1\sigma_z^2 \\ \Rightarrow \quad & \rho_{xz} = \operatorname{corr}(x_i, z_i) = \frac{\operatorname{cov}(x_i, z_i)}{\sigma_x\sigma_z} = \eta_1\frac{\sigma_z}{\sigma_x} \\ \Rightarrow \quad & \eta_1^2\sigma_z^2 = \rho_{xz}^2\sigma_x^2 \end{aligned}$$

One can therefore rewrite the formula for the variance of our IV estimator as

$$\sigma^2_{\text{IV}} = \frac{\delta^2\sigma^2_w + \sigma^2_u}{n\ \sigma^2_x\rho^2_{xz}} \tag{13}$$

These variance formula enable to retrieve the usual results regarding the determinant of the 2SLS variance as discussed in Hansen (2022) for instance. The variance of the estimator increases when the variance of the error σ^2_u increases. It decreases with the correlation between x and z, ρ_{xz} and with the variance of x, σ^2_x, for a given omitted variable.

II. 3 Comparison of efficiencies

The IV and naive regression variances are rather similar. Note that $\sigma^2_x\rho^2_{xz} < \sigma^2_x$ and,

$$\begin{aligned}
& \frac{\gamma^2\sigma^2_w}{\sigma^2_x} > 0 \\
\Leftrightarrow\quad & 1 - \frac{\gamma^2\sigma^2_w}{\sigma^2_x} < 1 \\
\Leftrightarrow\quad & \sigma^2_u + \delta^2\sigma^2_w\left(1 - \frac{\gamma^2\sigma^2_w}{\sigma^2_x}\right) < \sigma^2_u + \delta^2\sigma^2_w
\end{aligned}$$

Both these inequalities imply that $\boxed{\sigma^2_{\text{IV}} > \sigma^2_{\text{OVB}}}$. While unbiased, the IV estimator is less precise than the naive estimator. This is a well known and intuitive result that we recovered in a particular setting, tied to the modelling assumptions we made. This result is due to the fact that the IV only uses part of the variation in the endogenous dependent variable, the part that is explained by the instrument.

III Exaggeration ratio and its properties

Under construction.

References

Hansen, B. (2022), Econometrics, Princeton University Press, Princeton. 3, 8